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RL-Driven Carbon-Aware Serverless Scheduler

Project Proposal

Introduction

With the rapid growth of cloud computing, data centers consume significant amounts of electricity, contributing to increased carbon emissions. This project proposes an intelligent carbon-aware scheduling system that leverages Reinforcement Learning (RL) to optimize the execution of serverless workloads while minimizing environmental impact.

Problem Statement

Traditional cloud schedulers primarily focus on performance, cost, and resource utilization, often overlooking the carbon footprint associated with energy consumption. As a result, serverless tasks may be executed in regions powered by high-carbon energy sources, leading to unnecessary environmental impact.

Proposed Solution

The proposed RL-Driven Carbon-Aware Serverless Scheduler uses Reinforcement Learning to make intelligent decisions regarding where and when serverless functions should be executed. The system continuously monitors the carbon intensity of different cloud regions and selects locations that are currently powered by cleaner energy sources such as solar, wind, or hydroelectric power.

For non-time-sensitive workloads, the scheduler can also delay task execution until greener energy becomes available. By learning from historical and real-time energy data, the RL agent optimizes scheduling decisions while maintaining application performance and service quality.

Key Features

- Reinforcement Learning-based intelligent scheduling.
- Real-time carbon intensity monitoring across cloud regions.

- Dynamic selection of low-carbon execution locations.
- Delay-tolerant task scheduling for non-urgent workloads.
- Performance-aware optimization to maintain low latency and reliability.
- Reduced energy consumption and carbon emissions.

Expected Outcomes

The proposed system is expected to:

- Reduce the carbon footprint of cloud-based serverless applications.
- Improve energy efficiency across distributed cloud environments.
- Support sustainable and environmentally responsible computing practices.
- Maintain acceptable performance levels while optimizing for carbon reduction.

Conclusion

The RL-Driven Carbon-Aware Serverless Scheduler combines artificial intelligence and sustainable cloud computing to create a smarter and greener infrastructure. By intelligently scheduling serverless workloads based on carbon intensity and renewable energy availability, the system contributes to reducing environmental impact without compromising performance, making it a practical solution for future sustainable cloud platforms.